



PAK-AUSTRIA FACHHOCHSCHULE

INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY



Name : Hafsa Shahid

Reg no : B25F34CYS101

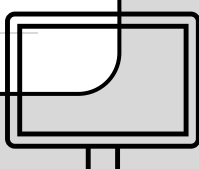
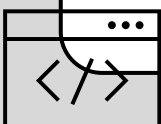
Program : Cybersecurity

Instructor's name: Hassan Ali

Date Submitted : 08 / 07 / 26

Deadline : 09 / 07 / 26

Assignment Report # 3





• INTRODUCTION •

In this class we learned Networking Fundamentals, IP Addressing, public and private IPs, NAT, Ports and Protocols, DNS and DHCP , and the differences between TCP and UDP.

We were given an assignment at the end; it goes as follows.

Assignments Tasks:

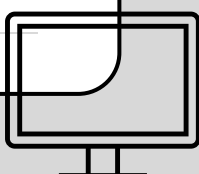
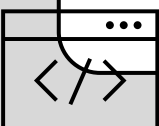
Task # 01 : *IP Address*, Find IP address.

Task # 02 : *Gateway*, Find Gateway.

Task # 03 : *DNS Server*, Find DNS Server.

Task # 04 : *Active Interfaces*, Find and List Active Interfaces.

Task # 05 : *Listening Ports*, Find and List Listening Ports.



• TASK # 01 •

Task # 01 : IP Address, Find IP address.

Steps Followed:

- Open Kali Linux, then open terminal by using shortcut Ctrl + Alt + t.
- Type “ip a” or “hostname -I” to get your IP address.

```
(kali@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:55:9a:b7 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.123.131/24 brd 192.168.123.255 scope global dynamic noprefixroute eth0  
        valid_lft 1261sec preferred_lft 1261sec  
    inet6 fe80::8de7:f8ab:7c0f:212e/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever
```

Task # 01: Command ip a

```
(kali@kali)-[~]  
$ hostname -I  
192.168.123.131
```

Task # 01: Command hostname -I

My Ip is **192.168.123.131**.

• TASK # 02 •

Task # 02 : Gateway, Find Gateway.

Steps Followed:

- Open Kali Linux, then open terminal by using shortcut Ctrl + Alt + t.
- Type “**ip route**” to get your Gateway.

```
(kali㉿kali)-[~]  
$ ip route  
default via 192.168.123.2 dev eth0 proto dhcp src 192.168.123.131 metric 100  
192.168.123.0/24 dev eth0 proto kernel scope link src 192.168.123.131 metric 100
```

Task # 02: Command ip route

My Gateway is **192.168.123.2**.

• TASK # 03 •

Task # 03 : DNS Server, Find DNS Server.

Steps Followed:

- Open Kali Linux, then open terminal by using shortcut Ctrl + Alt + t.
- Type “**hostname -i**” to get your DNS Server.

```
(kali㉿kali)-[~]  
$ hostname -i  
127.0.1.1
```

Task # 03: Command hostname -i

My DNS Server is **127.0.1.1**.

• TASK # 04 •

Task # 04 : Active Interfaces, Find and List Active Interfaces.

Steps Followed:

- Open Kali Linux, then open terminal by using shortcut Ctrl + Alt + t.
- Type “**ip a**” to list your Active Interfaces.

```
(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:55:9a:b7 brd ff:ff:ff:ff:ff:ff
    inet 192.168.123.131/24 brd 192.168.123.255 scope global dynamic noprefixroute eth0
        valid_lft 1261sec preferred_lft 1261sec
    inet6 fe80::8de7:f8ab:7c0f:212e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Task # 04: Command ip a

My Active Interfaces are **lo** and **eth0**.

• TASK # 05 •

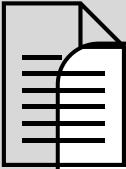
Task # 05 : Listening Ports, Find and List Listening Ports.

Steps Followed:

- Open Kali Linux, then open terminal by using shortcut Ctrl + Alt + t.
- Type “**sudo netstat -tulnup**” to list your Listening Ports.

```
(kali㉿kali)-[~]
$ sudo netstat -tulnup
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
```

Task # 05: Command sudo netstat -tulnup



As we can see, there are currently no Listening Ports. So, to force one, open another terminal and type “**python3 -m http.server 80**”.

```
(kali@kali)-[~]  
$ python3 -m http.server 80  
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
```

Task # 05: Command `python3 -m http.server 80`

Then go back to the first terminal and type the same command “**sudo netstat -tulnup**”.

```
(kali@kali)-[~]  
$ sudo netstat -tulnup  
Active Internet connections (only servers)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name  
tcp        0      0 0.0.0.0:80             0.0.0.0:*               LISTEN      75705/python3
```

Task # 05: Command `sudo netstat -tulnup`

Now we have a listening port.

Assignment Conclusion:

In this assignment we ran some commands in Kali Linux to get a better understanding of networking fundamentals. This increased and polished my ability to use and learn more about cybersecurity.

• The End •
